DevOps Labs Report

Git & GitHub | Docker Basics | Docker Advanced

1. Git & GitHub - branches, PRs, merge conflict

Repo: github.com/Kubepsychoctl/team-docs-warriors (3 PRs, 2 conflicts resolved)

2. Docker Basics (Week 1)

hello-world, nginx, ubuntu, own Dockerfile, Flask web app; image pushed to Docker Hub as kubernetesson/my-webapp:v1.0

3. Docker Advanced - Lab 1 (Week 4)

react-blog: multi-stage, alpine, non-root; 2.64 GB -> 191 MB (~14x smaller)

4. Docker Advanced - Lab 2 (Week 5)

shop-app: nginx + Flask + Redis on a custom network

5. Docker Practice - Lab 2: calculator (Flask)

Built, run and tested in the browser

6. Docker Practice - Lab 3a: QR generator (Node.js)

Built, run, QR generated, downloaded and decoded

7. Volumes - Lab 3b: PostgreSQL with persistent storage

Data survives container deletion; pg_dump and volume backups

8. Environment variables - Lab 4

config-app in development and production mode

9. Docker Compose - Lab 5

api + PostgreSQL + Redis + Nginx with healthchecks, networks, volumes

10. Part 10 - PaaS deployment on Render

Public HTTPS URL, /health, auto-deploy only after CI passes, failing commit not deployed

Appendix

Earlier local-simulation screenshots of the Git lab

Note: terminal screenshots are rendered from real command output; Docker was run in a sandbox (CA passed as build secret).

1. Git & GitHub lab

Git & GitHub

Pull requests #1-#3 (all merged) - real repo

GitHub PRs for Kubepsychoctl/team-docs-warriors (REST API)

```
$ gh api 'repos/Kubepsychoctl/team-docs-warriors/pulls?state=all' --jq '.[]|"#\(.number) \(.title) | merged=\
(.merged_at!=null) | \(.html_url)"'
#3 Add Carol: team contribution | merged=true | https://github.com/Kubepsychoctl/team-docs-warriors/pull/3
#2 Add Bob: team contribution | merged=true | https://github.com/Kubepsychoctl/team-docs-warriors/pull/2
#1 Add Alice: team contribution | merged=true | https://github.com/Kubepsychoctl/team-docs-warriors/pull/1
```

Git & GitHub

Real merge conflict in PR #2 (README.md)

Real merge conflict in Kubepsychoctl/team-docs-warriors (PR #2)

```
$ git merge main   # on feature/bob, main already contains PR #1
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

<<<<<<< HEAD
- Bob - CI/CD Engineer

## Technology Stack

- Jenkins
- Linux

## Setup Instructions

1. Install Jenkins
2. Configure pipeline
3. Run first build
=======
- Alice - DevOps Lead

## Technology Stack

- Docker
- Kubernetes

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
>>>>>>> ef2a46d
```

Git & GitHub

git log --oneline --graph --all and final README

Step 7 - git log --oneline --graph --all

```
$ git log --oneline --graph --all
*   c57b978 Merge pull request #3 from Kubepsychoctl/feature/carol
|\
| *   b1b5c9f Resolve merge conflict: combine team contributions
| |\
| |/
|/|
* |   72bad17 Merge pull request #2 from Kubepsychoctl/feature/bob
|\ \
| * \   f54c543 Resolve merge conflict: combine team contributions
| |\ \
| |/ /
|/| |
* | |   ef2a46d Merge pull request #1 from Kubepsychoctl/feature/alice
|\ \ \
| * | | 5414007 Add Alice: team info and setup instructions
|/ / /
| * / fbba07e Add Bob: team info and setup instructions
|/ /
| * bd0b03f Add Carol: team info and setup instructions
|/
* bd55fd3 Add README structure
* 451d473 Initial commit

$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer
- Carol - Cloud Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux
- Terraform
- AWS

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
7. Install Terraform
8. Run terraform init
9. Run terraform apply
```

2. Docker Basics

Docker Basics

Task 1 - hello-world

```
Task 1 - hello-world

$ docker --version
Docker version 29.9.2, build a82e528

$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
```

Docker Basics

Task 2 - nginx

```
Task 2 - nginx

$ docker run -d -p 8080:80 --name web nginx
Digest: sha256:5ab5b0b01097420873a3f3330f44248e34a3c2a3e8d5ee30658aeae8cb7435da
Status: Downloaded newer image for nginx:latest
866390c8209042a185258008b2a14f01a3386dacb2f69988c6f0ba82f28c0187

$ curl -s http://localhost:8080 | head -n 12
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>

$ docker logs web | tail -n 6
2026/11/03 16:04:23 [notice] 1#1: start worker processes
2026/11/03 16:04:23 [notice] 1#1: start worker process 20
2026/11/03 16:04:23 [notice] 1#1: start worker process 21
2026/11/03 16:04:23 [notice] 1#1: start worker process 22
2026/11/03 16:04:23 [notice] 1#1: start worker process 23
172.17.0.1 - - [03/Nov/2026:16:04:26 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.5.0" "-"

$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                  NAMES
866390c82090   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 3 seconds   0.0.0.0:8080->80/tcp   web

$ docker stop web && docker rm web
web
web
```

Docker Basics

Task 3 - ubuntu container

```
Task 3 - ubuntu container

$ docker run -it --name my-ubuntu ubuntu bash   # commands fed non-interactively
[os-release]
PRETTY_NAME="Ubuntu 26.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"

$ ls /
bin
boot
dev
etc
home
lib
lib64
media
mnt
opt
proc
root
run
sbin
srv
sys
tmp
usr
var
Hello

$ docker rm my-ubuntu
my-ubuntu
```

Docker Basics

Task 4 - own Dockerfile (hello-docker)

Task 4 - own Dockerfile

```
$ cd hello-docker && docker build -t hello-docker .
#7 DONE 0.0s

#8 exporting to image
#8 exporting layers 0.1s done
#8 exporting manifest sha256:e6202534159b2c3490837e0ddf8ddc0b1e58cad884ffcee6c5c0af0091baba42 done
#8 exporting config sha256:33918b8c39c632e92a773d62bd9972a076da852ebfdc741e8d076b853cff5a7e done
#8 exporting attestation manifest sha256:be1f8fccd53d4d012bc90bd5c4c630c104b49ed637a2e644e5570b89134c5a11 0.0s done
#8 exporting manifest list sha256:415be81095c9c0ed64544e2c64cda080c057b7e412901f51958af0a651a84d3f done
#8 naming to docker.io/library/hello-docker:latest done
#8 unpacking to docker.io/library/hello-docker:latest
#8 unpacking to docker.io/library/hello-docker:latest 1.5s done
#8 DONE 1.6s

$ docker run --rm hello-docker
==================================================
Привет из Docker!
Моё имя: Kubepsychoctl
Этот код работает в контейнере!
==================================================
```

Docker Basics

Lab - Flask my-webapp build and run

Lab - Flask my-webapp

```
$ docker build -t my-webapp .
#3 DONE 0.0s
#4 DONE 0.0s
#5 DONE 0.0s
#8 6.567 Successfully installed Flask-3.0.0 Jinja2-3.1.6 MarkupSafe-3.0.4 Werkzeug-3.1.9 blinker-1.9.0 click-8.5.0
itsdangerous-2.2.0
#8 DONE 6.9s
#9 DONE 0.0s
#10 naming to docker.io/library/my-webapp:latest done
#10 DONE 1.2s

$ docker run -d -p 5000:5000 --name webapp my-webapp
8d35399b9f3e8fa04ac595edac323d187aed36182f5f9d5a04d7e877d08393fb

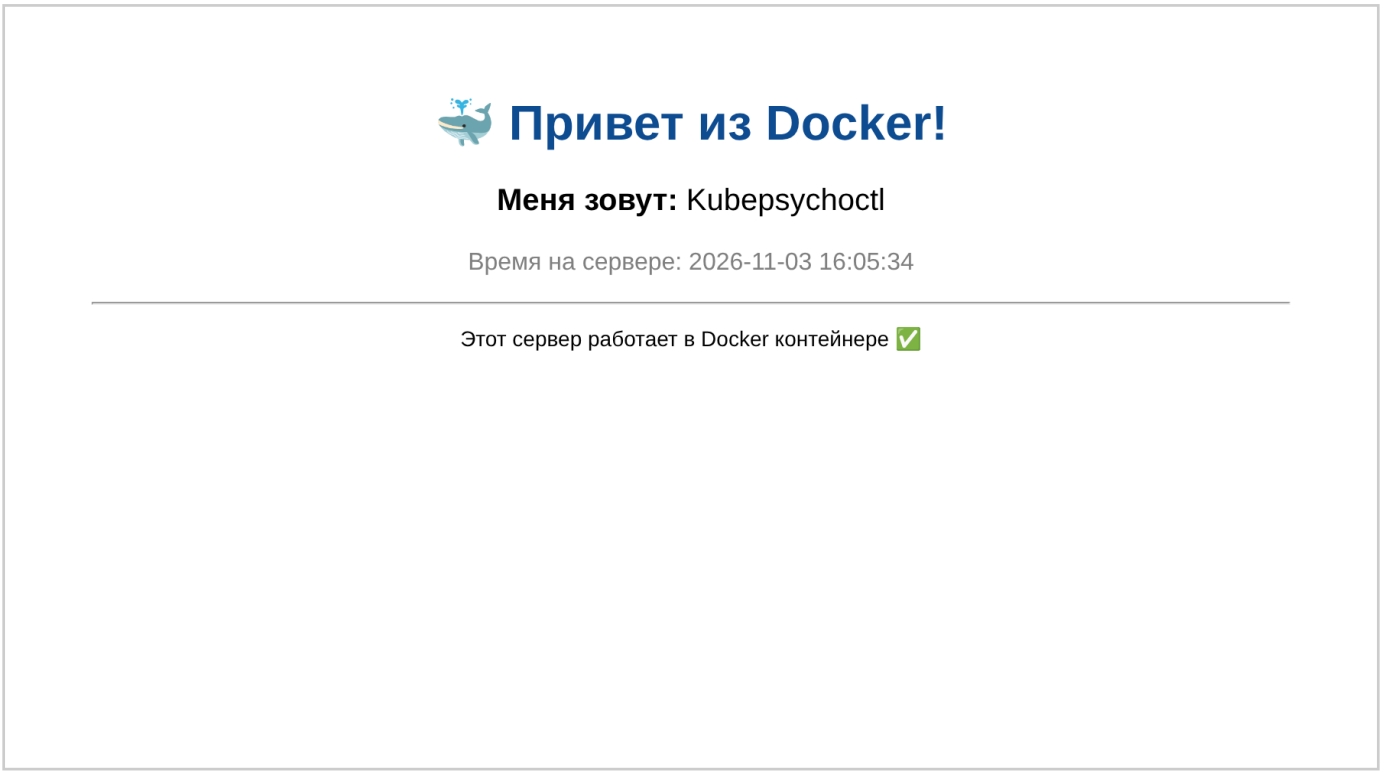
$ curl -s http://localhost:5000

    <html>
    <body style="font-family: Arial; text-align: center; padding: 50px;">
        <h1 style="color: #0E4C92;">🐳 Привет из Docker!</h1>
        <p style="font-size: 20px;">
            <strong>Меня зовут:</strong> Kubepsychoctl
        </p>
        <p style="color: gray;">
            Время на сервере: 2026-11-03 16:05:21
        </p>
        <hr>
        <p style="font-size: 14px;">
            Этот сервер работает в Docker контейнере ✅
        </p>
    </body>
    </html>

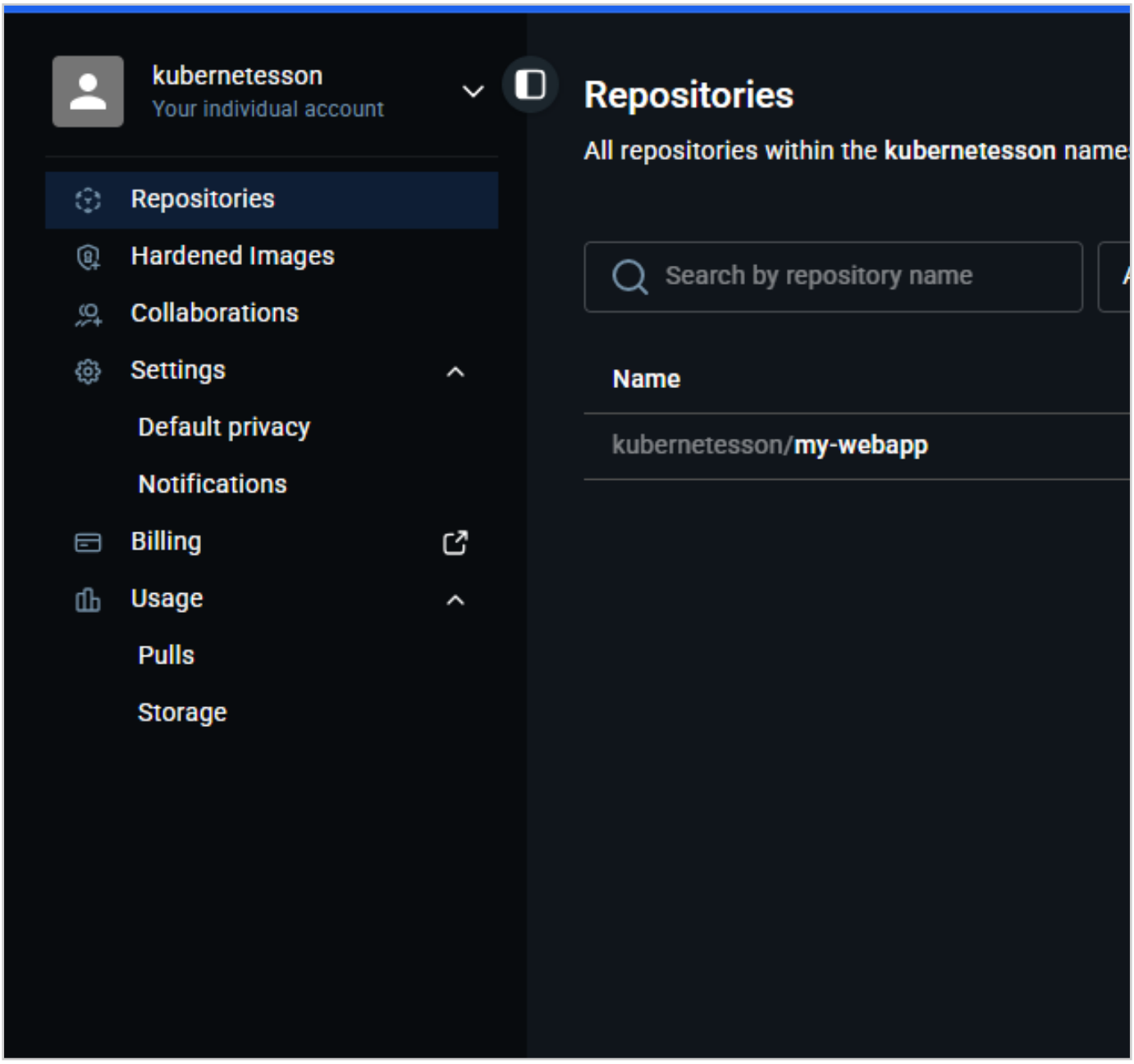
$ docker ps
CONTAINER ID   IMAGE       COMMAND           CREATED         STATUS         PORTS                    NAMES
8d35399b9f3e   my-webapp   "python app.py"   4 seconds ago   Up 4 seconds   0.0.0.0:5000->5000/tcp   webapp

$ docker images
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-docker:latest  415be81095c9        187MB         45.6MB
hello-world:latest   c567b231f656       25.9kB         9.49kB   U
my-webapp:latest     d1d77635a9a5        211MB         51.3MB   U
nginx:latest         5ab5b0b01097        243MB         66.8MB
ubuntu:latest        6a06111c3b0a        163MB         45.9MB
```

my-webapp in the browser



Docker Hub account kubernetesson: repository my-webapp (pushed as v1.0)



3. Docker Advanced - Lab 1

Docker Advanced - Lab 1 (react-blog)

Image size: naive 2.64 GB vs optimized 191 MB

```
Lab 1 - image size: naive vs optimized (multi-stage)

$ docker images | grep react-blog
IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
react-blog-naive:latest   eda256c34f8b       2.64GB          561MB
react-blog:latest         0a3730d56616        191MB         46.8MB   U

naive:     node:18, COPY . ., no .dockerignore, single stage = 2.64 GB
optimized: multi-stage + alpine + non-root + .dockerignore = 191 MB (~14x smaller, < 200 MB)
```

Docker Advanced - Lab 1 (react-blog)

Run, non-root user, health check, layers

```
Lab 1 - run, non-root user, health check

$ docker images | grep react-blog
IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
react-blog-naive:latest   eda256c34f8b       2.64GB          561MB
react-blog:latest         0a3730d56616        191MB         46.8MB

$ docker run -d -p 3000:3000 --name blog react-blog
135f873347e1114e315473d894bf451b19f9133ff60b02552cc6061718aab9bc

$ docker exec blog whoami   # non-root check
appuser
uid=1001(appuser) gid=1001(appgroup) groups=1001(appgroup)

$ curl -s localhost:3000/api/health
{"status":"ok","message":"Server is running"}

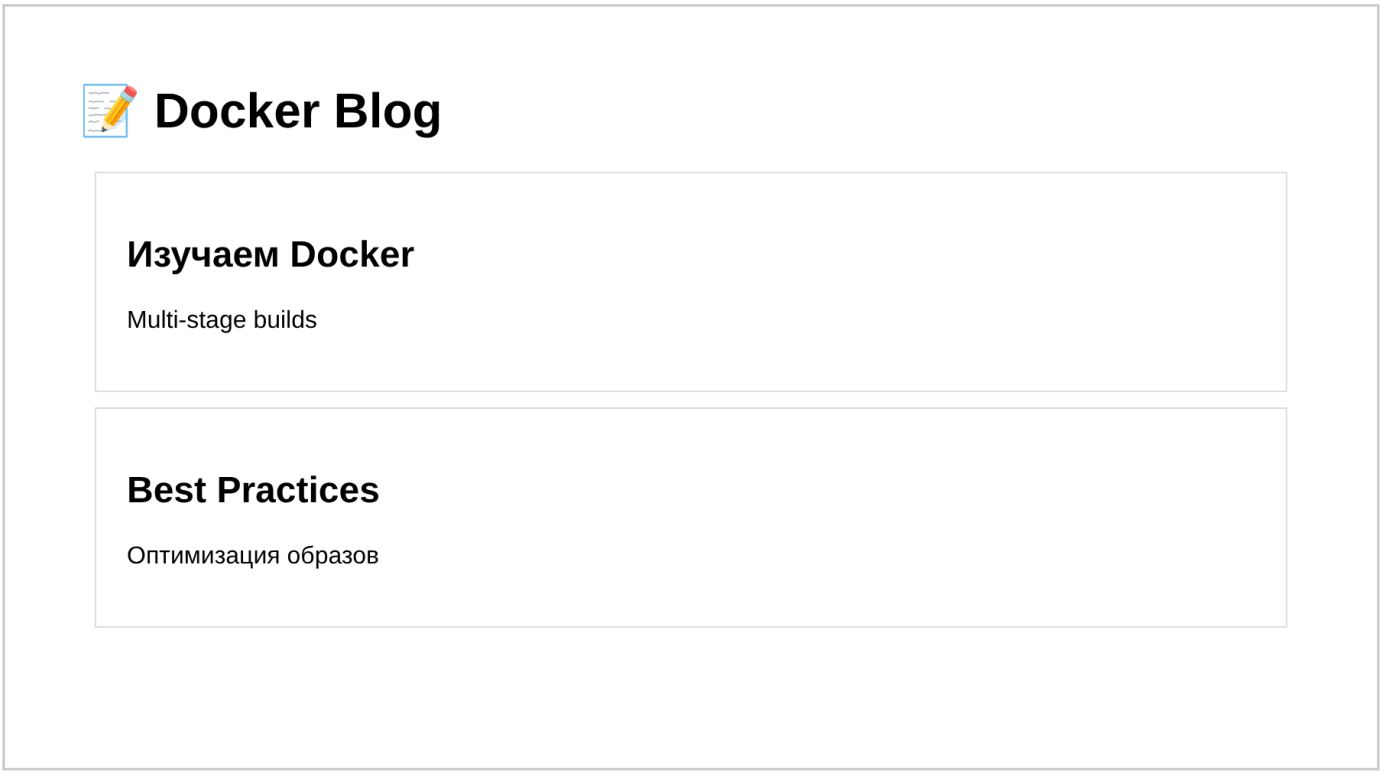
$ curl -s localhost:3000 | head -c 250
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" cont
ent="width=device-width,initial-scale=1"/><meta name="theme-color" content="#000000"/><meta name="description" content="Web
si

$ docker logs blog
Server running on port 3000

$ docker history react-blog --format "{{.Size}}\t{{.CreatedBy}}" | head -n 8
0B      CMD ["node" "server.js"]
0B      EXPOSE [3000/tcp]
0B      USER appuser
1.39MB  COPY /app/client/build ./build # buildkit
12.3kB  COPY server/server.js . # buildkit
6.99MB  RUN /bin/sh -c NODE_EXTRA_CA_CERTS=/tmp/ca.c…
45.1kB  COPY server/package*.json ./ # buildkit
8.19kB  WORKDIR /app
```

Docker Advanced - Lab 1 (react-blog)

react-blog in the browser



4. Docker Advanced - Lab 2

Docker Advanced - Lab 2 (shop-app)

./start.sh - network and 3 containers

Lab 2 - ./start.sh (network + 3 containers)

```
$ ./start.sh
a689950e4b8f621e1007e8d4c4b0783090bb5f03bb34e78dca9b45785af5f9f0
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
e25d07126ba7: Pulling fs layer
cef7ad5a05c6: Pulling fs layer
759efa4719c9: Pulling fs layer
a239ec2b8fe1: Pulling fs layer
ae75cb664c8f: Pulling fs layer
1ea868e596ec: Pulling fs layer
4f4fb700ef54: Pulling fs layer
4f4fb700ef54: Already exists
b5e4a0ae9c7c: Download complete
4d584b19579f: Download complete
cef7ad5a05c6: Download complete
759efa4719c9: Download complete
e25d07126ba7: Download complete
a239ec2b8fe1: Download complete
ae75cb664c8f: Download complete
759efa4719c9: Pull complete
a239ec2b8fe1: Pull complete
cef7ad5a05c6: Pull complete
1ea868e596ec: Download complete
ae75cb664c8f: Pull complete
1ea868e596ec: Pull complete
4f4fb700ef54: Pull complete
e25d07126ba7: Pull complete
Digest: sha256:8da19c7399dca900c6822a1cf30f9d2285585e2c4b457ff4095b0f4500054d3f
Status: Downloaded newer image for redis:7-alpine
08073f7b7955548ca2e0389eb25d16eee0a13efbbdb19c4a03bdd193d7de8225
95371a2c83d9a27cd686396c1050fcd1fc2a1842f3d21c0e6f021e1ab6f8f879
96a914695f25cf27b49dc447f0df87db544c08cfb61313c41198ea54a0fcbfba
Shop запущен!
Frontend: http://localhost:8080
Backend API: http://localhost:5000/products
exit=0
```

Docker Advanced - Lab 2 (shop-app)

Network, DNS by name, API, Redis

```
Lab 2 - network, DNS by name, API, Redis

$ docker ps
NAMES      IMAGE            PORTS                    STATUS
frontend   shop-frontend    0.0.0.0:8080->80/tcp     Up 5 seconds
backend    shop-backend     0.0.0.0:5000->5000/tcp   Up 10 seconds
redis      redis:7-alpine   6379/tcp                 Up 16 seconds

$ docker network ls | grep shop
NETWORK ID     NAME           DRIVER    SCOPE
a689950e4b8f   shop-network   bridge    local

$ docker network inspect shop-network --format "{{range .Containers}}{{.Name}} {{.IPv4Address}}{{println}}{{end}}"
redis 172.18.0.2/16
backend 172.18.0.3/16
frontend 172.18.0.4/16

$ docker exec backend python -c "import socket;print(socket.gethostbyname(\"redis\"))"   # DNS by name
redis -> 172.18.0.2

$ curl -s localhost:5000/products
[{"id":1,"name":"Docker T-Shirt","price":25},{"id":2,"name":"Kubernetes Mug","price":15},{"id":3,"name":"DevOps Book","pric
e":40}]

$ curl -s -X POST localhost:5000/cart/add/1 ; ...add/3
{"success":true}
{"success":true}

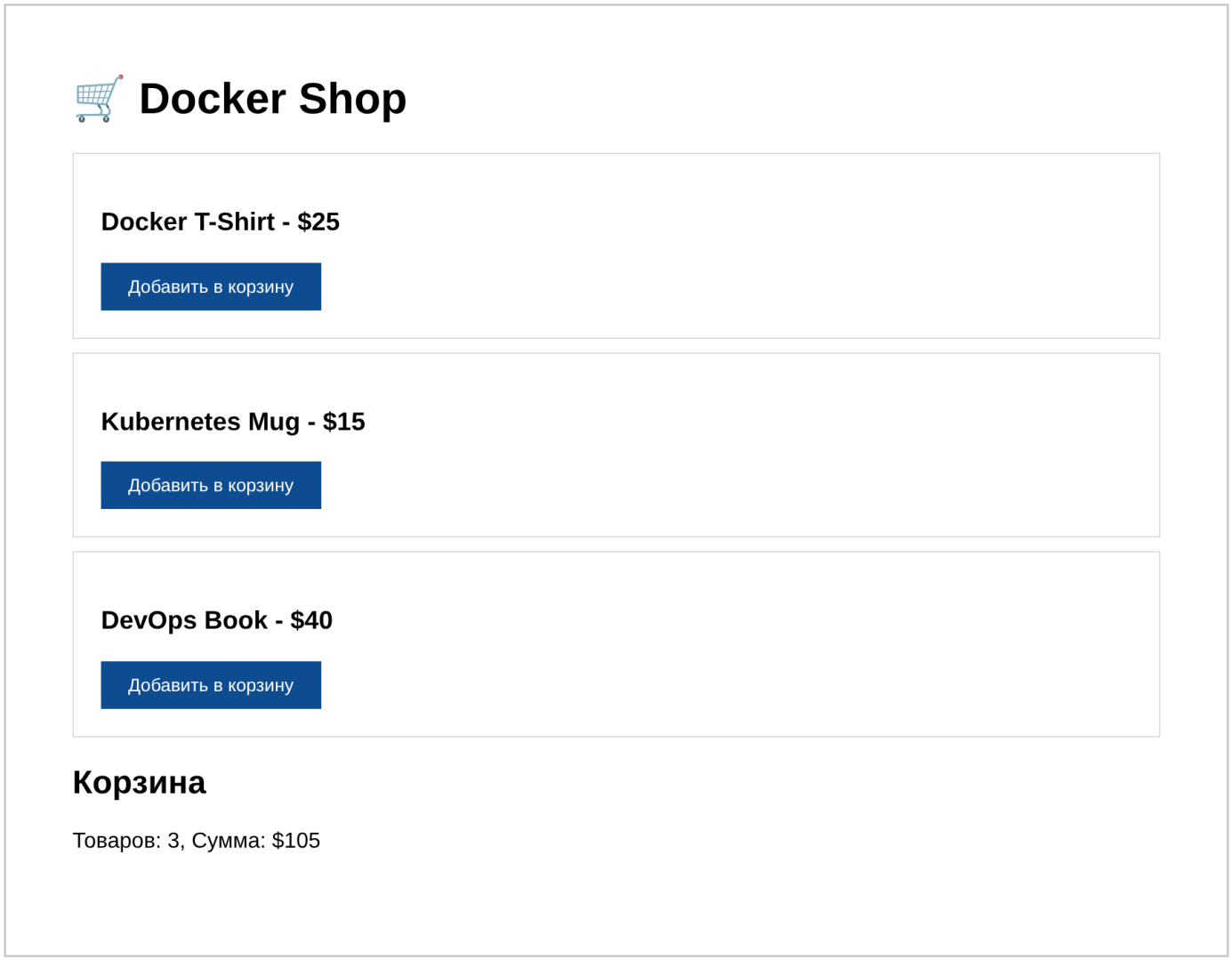
$ curl -s localhost:5000/cart
{"count":2,"items":[{"id":3,"name":"DevOps Book","price":40},{"id":1,"name":"Docker T-Shirt","price":25}],"total":65}

$ docker exec redis redis-cli LLEN cart
2

$ curl -s localhost:8080 | head -n 4
<!DOCTYPE html>
<html>
<head>
  <title>Docker Shop</title>
```

Docker Advanced - Lab 2 (shop-app)

Docker Shop with cart in the browser



5. Docker Practice - Lab 2: Calculator

- Flask web calculator (add, subtract, multiply, divide) packaged with the Dockerfile from the handout.
- Image calculator-app built and started with: docker run -d -p 5000:5000 --name calc calculator-app.
- Tested in a browser: 12.5 x 4 = 50.0; 5 / 0 shows the message 'Ошибка: деление на 0'. Also 6 x 7 = 42.0 via curl.
- Step 7 (Docker Hub): kubernetesson/calculator-app:v1.0 pushed, see the Docker Hub section.

Practice - Lab 2 (calculator)

Build and run

Lab 2 - Calculator (Flask): build and run

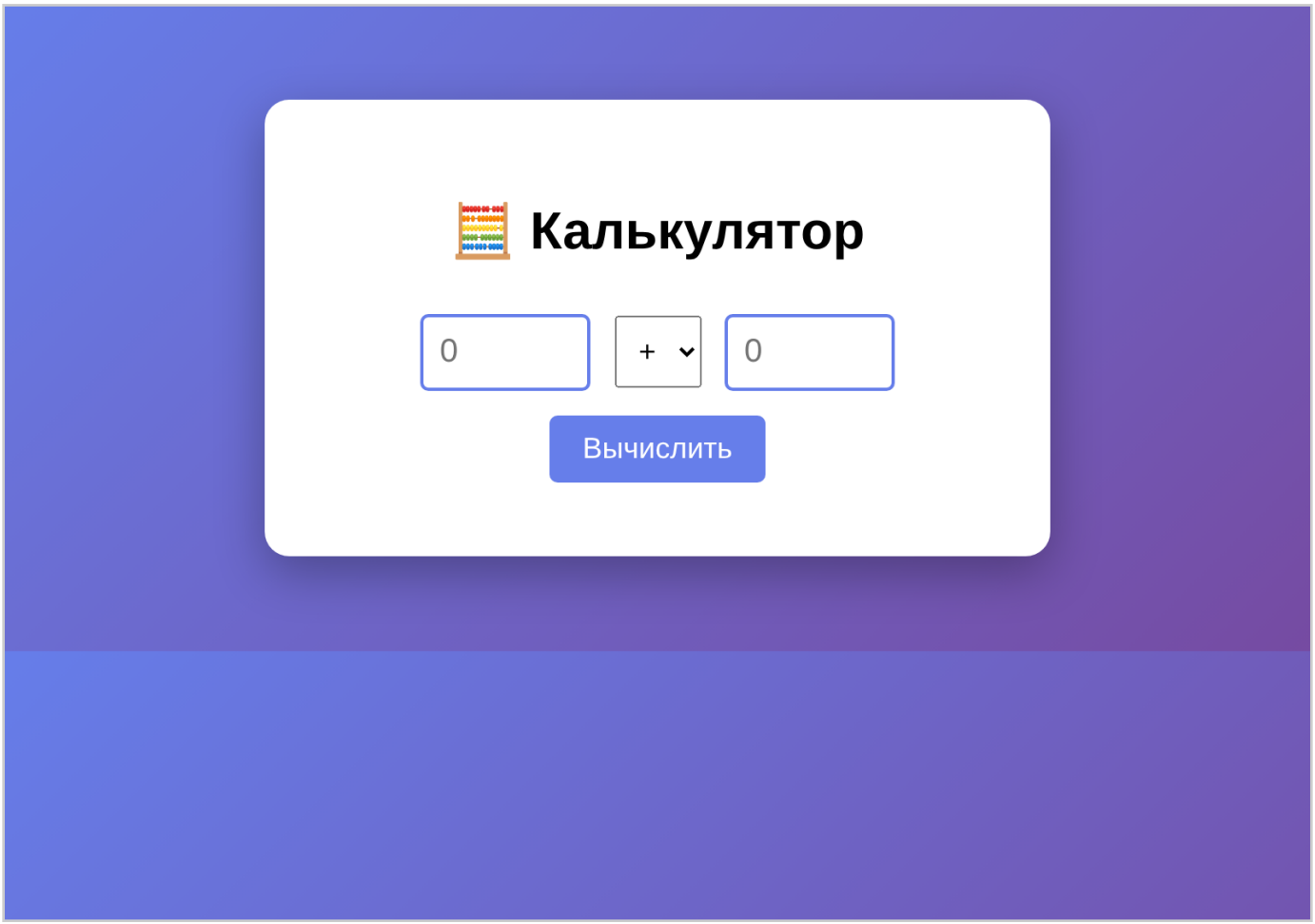
```
$ docker build -t calculator-app .
#10 naming to docker.io/library/calculator-app:latest done

$ docker run -d -p 5000:5000 --name calc calculator-app
13f985fc241cc1f2790b659bf5c7282b25d8fac63776523548f8b8bfe7c653cf

$ curl -s -X POST localhost:5000 -d 'num1=6&operation=multiply&num2=7' | grep result
    <div class="result">Результат: 42.0</div>
```

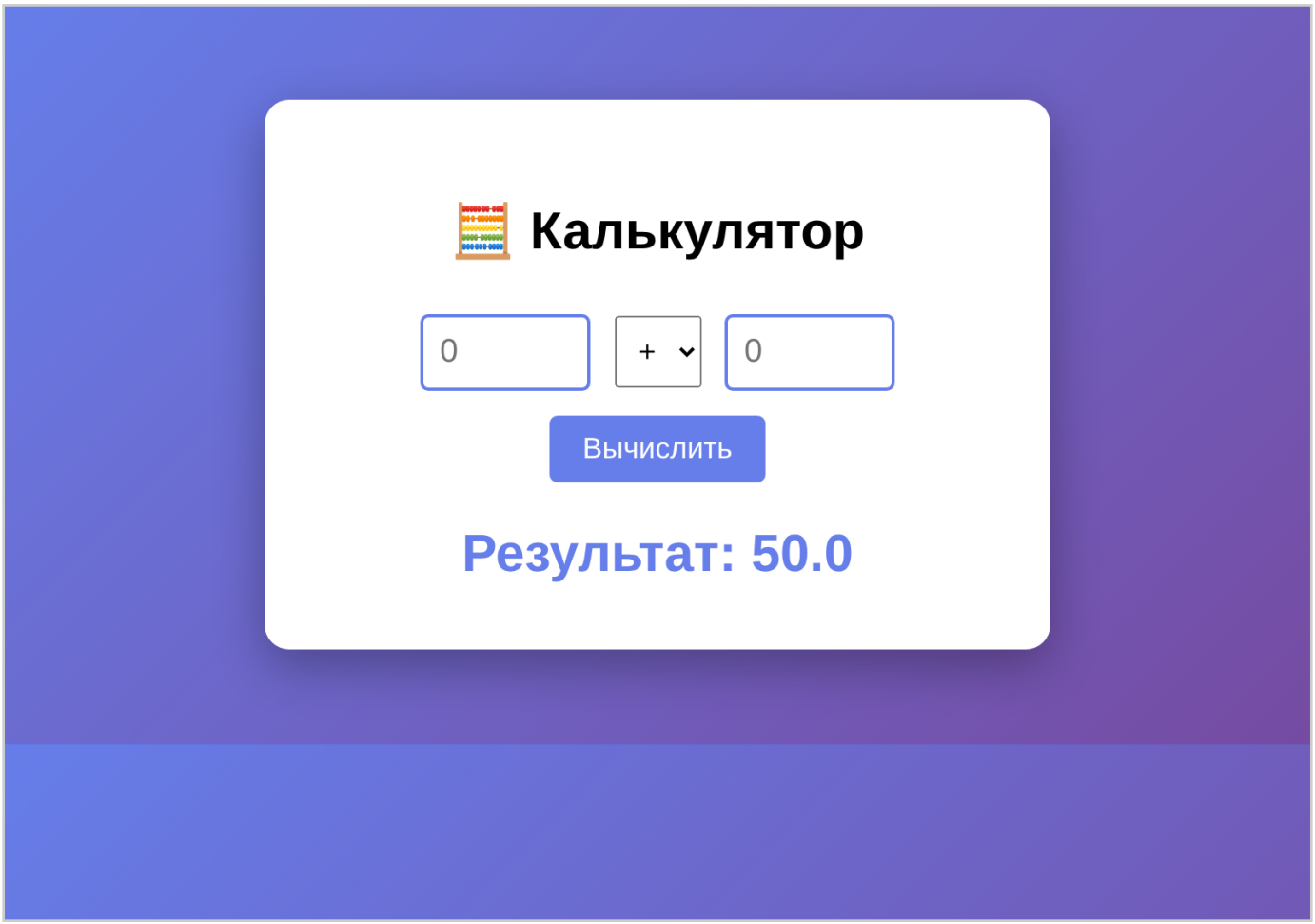
Practice - Lab 2 (calculator)

Calculator in the browser



Practice - Lab 2 (calculator)

Result of 12.5 x 4



6. Docker Practice - Lab 3a: QR generator

- Node.js + Express + qrcode application, node:18-alpine image, Dockerfile from the handout.
- Built as qr-generator and started on port 3000. Text 'Hello Docker' produced a QR code; the 'Скачать QR-код' button downloaded qrcode.png.
- The downloaded PNG was decoded with OpenCV and returned 'Hello Docker', so the QR code is valid.
- Step 7 (Docker Hub): kubernetesson/qr-generator:v1.0 pushed, see the Docker Hub section.

Practice - Lab 3a (QR generator)

Build and run

Lab 3a - QR generator (Node.js): build and run

```
$ docker build -t qr-generator .
#8 7.745 added 97 packages, and audited 98 packages in 4s
#10 naming to docker.io/library/qr-generator:latest done

$ docker run -d -p 3000:3000 --name qr qr-generator
cf1bdeb02f54276513a15cbf98cfa3e827a5b83d15f8993b4bd8bbbfb9d10a5f

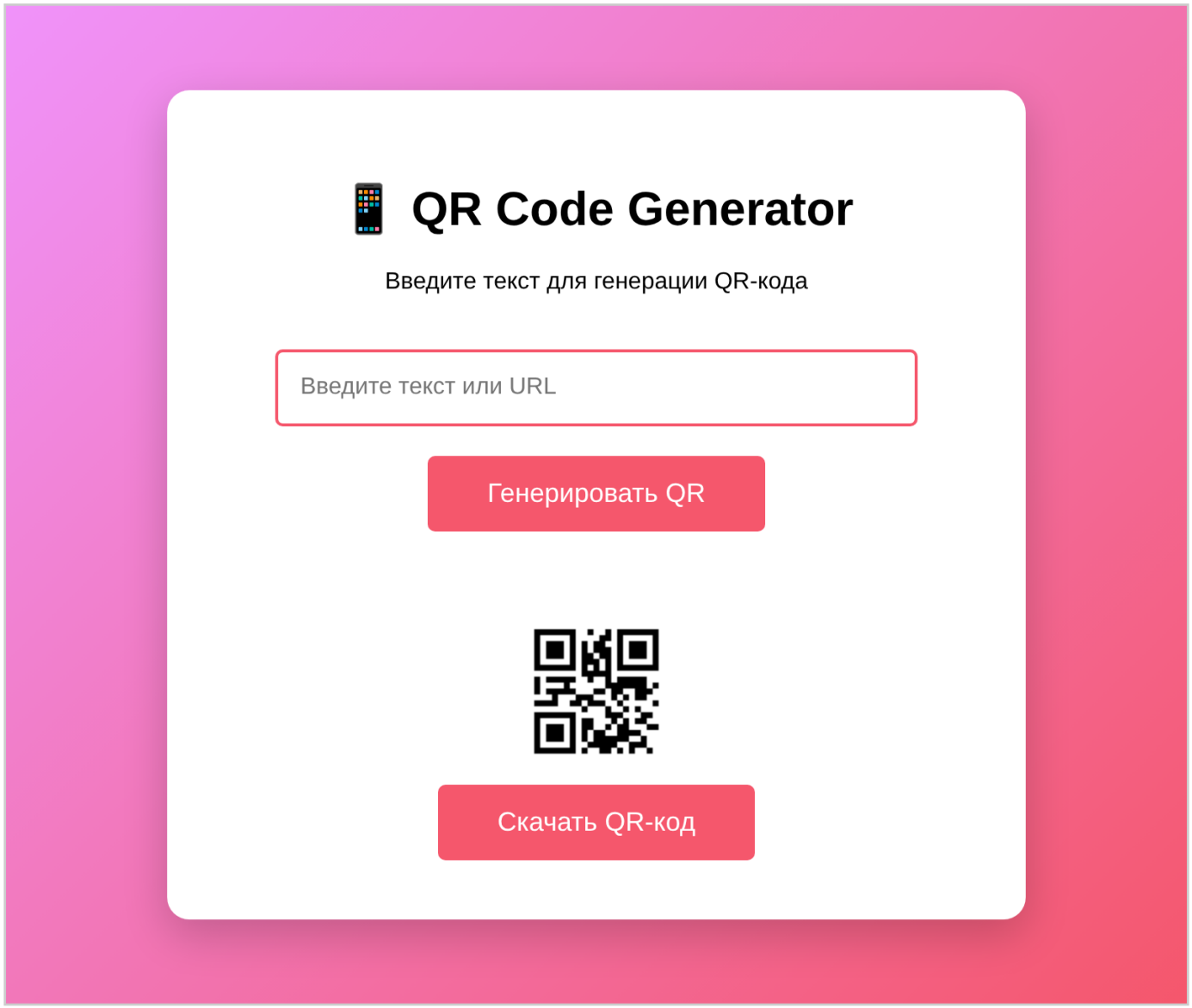
$ docker logs qr
> qr-generator@1.0.0 start
> node server.js

Сервер запущен на http://localhost:3000

# The downloaded qrcode.png (116x116 PNG) was decoded back with OpenCV:
decoded QR: 'Hello Docker'
```

Practice - Lab 3a (QR generator)

QR code generated in the browser



Docker Hub - Labs 2 and 3a (step 7)

- Both images were published to Docker Hub account kubernetesson: kubernetesson/calculator-app:v1.0 and kubernetesson/qr-generator:v1.0.
- Digests: calculator-app sha256:b97903940fd8842317b6ad1d719a5fbc66f7ea8368e8fec1da21c11bba60f443; qr-generator sha256:1f50a2bec7133750e1ce5fab8e52451312a0b4eb2dddd788193a6dcd67503e1c.
- Verified by deleting the local copies, pulling from Docker Hub and running: the calculator returns 6 x 7 = 42.0 and the QR generator returns a QR image.
- Public pages: https://hub.docker.com/r/kubernetesson/calculator-app and https://hub.docker.com/r/kubernetesson/qr-generator

Docker Hub (Labs 2 and 3a)

docker login, tag, push

Docker Hub - push calculator-app and qr-generator (account kubernetesson)

```
$ docker login -u kubernetesson --password-stdin
Login Succeeded

$ docker tag calculator-app kubernetesson/calculator-app:v1.0
$ docker push kubernetesson/calculator-app:v1.0
The push refers to repository [docker.io/kubernetesson/calculator-app]
6fc242d221a6: Mounted from library/nginx
3c0c4313fac2: Pushed
ee5aaeba9945: Pushed
44136fa355b3: Pushed
b39b2e52f906: Pushed
361db8171547: Pushed
636ec4f9c923: Pushed
90417d3f590a: Pushed
24ff04c92107: Pushed
f28be0661c5f: Pushed
v1.0: digest: sha256:b97903940fd8842317b6ad1d719a5fbc66f7ea8368e8fec1da21c11bba60f443 size: 856

$ docker tag qr-generator kubernetesson/qr-generator:v1.0
$ docker push kubernetesson/qr-generator:v1.0
The push refers to repository [docker.io/kubernetesson/qr-generator]
813c7167225f: Pushed
f18232174bc9: Pushed
44136fa355b3: Mounted from kubernetesson/calculator-app
25ff2da83641: Pushed
1e5a4c89cee5: Pushed
5d9ba382c10d: Pushed
cde6ff73ce38: Pushed
c17616436cf9: Pushed
9eddea442117: Pushed
dd71dde834b5: Pushed
v1.0: digest: sha256:1f50a2bec7133750e1ce5fab8e52451312a0b4eb2dddd788193a6dcd67503e1c size: 856
```

Docker Hub (Labs 2 and 3a)

pull back from Docker Hub and run

Docker Hub - pull the images back and run them

```
$ docker rmi kubernetesson/calculator-app:v1.0 kubernetesson/qr-generator:v1.0   # remove local copies
untagged: 2

$ docker pull kubernetesson/calculator-app:v1.0
Digest: sha256:b97903940fd8842317b6ad1d719a5fbc66f7ea8368e8fec1da21c11bba60f443
Status: Downloaded newer image for kubernetesson/calculator-app:v1.0

$ docker pull kubernetesson/qr-generator:v1.0
Digest: sha256:1f50a2bec7133750e1ce5fab8e52451312a0b4eb2dddd788193a6dcd67503e1c
Status: Downloaded newer image for kubernetesson/qr-generator:v1.0

$ docker run -d -p 5000:5000 --name v-calculator-app kubernetesson/calculator-app:v1.0
09cf1a3d1cd6

$ curl -s -X POST localhost:5000 -d "num1=6&operation=multiply&num2=7" | grep result
    .result {
      <div class="result">Результат: 42.0</div>

$ docker ps
NAMES              IMAGE                                 PORTS                    STATUS
v-calculator-app   kubernetesson/calculator-app:v1.0   0.0.0.0:5000->5000/tcp   Up 4 seconds
v-qr-generator     kubernetesson/qr-generator:v1.0     0.0.0.0:3000->3000/tcp   Up 25 seconds

$ docker run -d -p 3000:3000 --name v-qr-generator kubernetesson/qr-generator:v1.0
$ curl -s -X POST localhost:3000/generate -d "text=Hello Docker" | grep -c "data:image/png"
2   (QR image present in the response)

$ docker logout
Removing login credentials for https://index.docker.io/v1/
```

7. Volumes - Lab 3b: PostgreSQL with persistent

- Named volume postgres-lab-data mounted into postgres:15-alpine with --mount source=...,target=/var/lib/postgresql/data.
- init.sql created tables students and courses (4 + 3 rows).
- Container postgres-lab was stopped and deleted; the volume remained. A new container postgres-lab-2 with the same volume returned all 4 students.
- Backups: pg_dump -> backup.sql (4.1 KB) and tar czf of the volume -> postgres-volume-backup.tar.gz (6.4 MB).

Volumes - Lab 3b

Volume, PostgreSQL, init.sql

Lab 3b - PostgreSQL + volume: create, init, data

```
$ docker volume create postgres-lab-data
postgres-lab-data
$ docker run -d --name postgres-lab --mount source=postgres-lab-data,target=/var/lib/postgresql/data -e POSTGRES_USER=studen
t -e POSTGRES_PASSWORD=student123 -e POSTGRES_DB=university -p 5432:5432 postgres:15-alpine
e2a0989faff5195f3de52c75cea2535844dc828a282a82b72892b4a1d51608ff
$ docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED              STATUS                  PORTS
NAMES
e2a0989faff5   postgres:15-alpine   "docker-entrypoint.s…"   1 second ago         Up Less than a second   0.0.0.0:5432->543
2/tcp   postgres-lab
cf1bdeb02f54   qr-generator         "docker-entrypoint.s…"   49 seconds ago       Up 48 seconds           0.0.0.0:3000->300
0/tcp   qr
13f985fc241c   calculator-app       "python app.py"          About a minute ago   Up About a minute       0.0.0.0:5000->500
0/tcp   calc
$ docker logs postgres-lab
$ docker exec -i postgres-lab psql -U student -d university < init.sql
CREATE TABLE
CREATE TABLE
INSERT 0 4
INSERT 0 3
$ docker exec postgres-lab psql -U student -d university -c 'SELECT * FROM students;'
 id |     name      |        email         |      major       | year |        created_at
----+---------------+----------------------+------------------+------+--------------------------
  1 | Alice Johnson | alice@university.edu | Computer Science |    3 | 2026-11-04 06:23:28.78091
  2 | Bob Smith     | bob@university.edu   | Mathematics      |    2 | 2026-11-04 06:23:28.78091
  3 | Carol White   | carol@university.edu | Physics          |    4 | 2026-11-04 06:23:28.78091
  4 | David Brown   | david@university.edu | Computer Science |    1 | 2026-11-04 06:23:28.78091
(4 rows)

$ docker exec postgres-lab psql -U student -d university -c 'SELECT * FROM courses;'
 id |      title       | credits |  instructor
----+------------------+---------+--------------
  1 | Database Systems |       4 | Dr. Wilson
  2 | Algorithms       |       3 | Dr. Martinez
  3 | Web Development  |       3 | Dr. Anderson
(3 rows)

$ docker exec postgres-lab psql -U student -d university -c '\dt'
          List of relations
 Schema |   Name    | Type  |  Owner
--------+-----------+-------+---------
 public | courses   | table | student
 public | students  | table | student
(2 rows)
```

Volumes - Lab 3b

Persistence test and backups

Lab 3b - delete container, new container, same data; backups

```
$ docker stop postgres-lab
postgres-lab
$ docker rm postgres-lab
postgres-lab
$ docker volume ls
local     postgres-lab-data
$ docker volume inspect postgres-lab-data
[
    {
        "CreatedAt": "2026-11-04T06:23:26Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/postgres-lab-data/_data",
        "Name": "postgres-lab-data",
        "Options": null,
        "Scope": "local"
    }
]
$ docker run -d --name postgres-lab-2 --mount source=postgres-lab-data,target=/var/lib/postgresql/data -e POSTGRES_USER=stud
ent -e POSTGRES_PASSWORD=student123 -p 5432:5432 postgres:15-alpine
7fcd1c865c5c2f19e00edfb92087a049cb5649869ba63c16f9b8e2b6f1c35478
$ docker exec postgres-lab-2 psql -U student -d university -c 'SELECT * FROM students;'
 id |     name      |        email         |      major       | year |        created_at
----+---------------+----------------------+------------------+------+--------------------------
  1 | Alice Johnson | alice@university.edu | Computer Science |    3 | 2026-11-04 06:23:28.78091
  2 | Bob Smith     | bob@university.edu   | Mathematics      |    2 | 2026-11-04 06:23:28.78091
  3 | Carol White   | carol@university.edu | Physics          |    4 | 2026-11-04 06:23:28.78091
  4 | David Brown   | david@university.edu | Computer Science |    1 | 2026-11-04 06:23:28.78091
(4 rows)

$ docker exec postgres-lab-2 pg_dump -U student university
$ docker run --rm --mount source=postgres-lab-data,target=/data -v /home/user/team-docs-warriors/docker-lab/practice/lab3-vo
lumes:/backup ubuntu tar czf /backup/postgres-volume-backup.tar.gz /data
tar: Removing leading `/' from member names
$ ls -lh backup.sql postgres-volume-backup.tar.gz
-rw-r--r-- 1 root root 4.1K Nov  4 06:23 backup.sql
-rw-r--r-- 1 root root 6.4M Nov  4 06:23 postgres-volume-backup.tar.gz
```

8. Environment variables - Lab 4

- Flask app configured only via environment variables (config.py), separate .env.development and .env.production, .env.example for the team, .gitignore for secrets.
- The Dockerfile contains no secrets and does not copy .env files (ls -a /app shows only app.py, config.py, requirements.txt); the app runs as non-root appuser.
- Without the variables the container exits at startup (KeyError: SECRET_KEY). Development: --env-file .env.development, /db-test succeeded. Production: custom network app-network, DB_HOST=prod-postgres, port 5001, /db-test succeeded.
- On Linux host.docker.internal needs --add-host host.docker.internal:host-gateway (used for the development run).

Environment variables - Lab 4

Build, image contents, validation

```
Lab 4 - build, no .env in image, non-root, validation without env

# Part 4.1 - PostgreSQL for dev testing
$ docker run -d --name dev-postgres -e POSTGRES_USER=devuser -e POSTGRES_PASSWORD=devpassword123 -e POSTGRES_DB=devdb -p 543
2:5432 postgres:15-alpine
62bc738e0d817f03fed65af05ca4b4bf480418ca51e8028e33faaab281e0c59b

# Part 4.2 - build image
$ docker build -t config-app .
#11 naming to docker.io/library/config-app:latest done

# the image must not contain .env files:
$ docker run --rm --entrypoint sh config-app -c ls -a /app; echo; echo whoami: $(whoami)
.
..
app.py
config.py
requirements.txt

whoami: appuser

# Without env vars the app refuses to start (validation):
$ docker run --rm config-app
Traceback (most recent call last):
  File "/app/app.py", line 2, in <module>
    from config import Config
  File "/app/config.py", line 4, in <module>
    class Config:
  File "/app/config.py", line 6, in Config
    SECRET_KEY = os.environ["SECRET_KEY"]
                 ~~~~~~~~~~^^^^^^^^^^^^^^
  File "<frozen os>", line 679, in __getitem__
KeyError: 'SECRET_KEY'
```

Environment variables - Lab 4

Development mode

Lab 4 - development mode (--env-file .env.development)

```
# Part 4.3 - development mode
$ docker run -d --name config-app-dev --env-file .env.development --add-host host.docker.internal:host-gateway -p 5000:5000
config-app
741e9867110f8b77c8eeec11975010ae7b11fd0d4fd3a31e2631d07961a79d7a

$ curl -s http://localhost:5000/
{
  "database_host": "host.docker.internal",
  "debug": true,
  "environment": "development",
  "message": "Config API"
}

$ curl -s http://localhost:5000/db-test
{
  "database": "PostgreSQL 15.19 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit",
  "status": "success"
}

$ docker logs config-app-dev
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.3:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 380-157-241
172.17.0.1 - - [04/Nov/2026 06:24:13] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [04/Nov/2026 06:24:13] "GET /db-test HTTP/1.1" 200 -

# env vars inside the container (secrets hidden):
$ sh -c docker exec config-app-dev env | grep -E '^(APP_ENV|DEBUG|DB_HOST|DB_NAME|DB_USER|FLASK_APP)=' | sort
APP_ENV=development
DB_HOST=host.docker.internal
DB_NAME=devdb
DB_USER=devuser
DEBUG=true
FLASK_APP=app.py

# override one variable on the command line (-e wins over env-file):
$ sh -c docker run --rm --env-file .env.development -e DEBUG=false --entrypoint python config-app -c 'from config import Con
fig; print("DEBUG =", Config.DEBUG, "| APP_ENV =", Config.APP_ENV)'
DEBUG = False | APP_ENV = development
```

Environment variables - Lab 4

Production mode (custom network)

Lab 4 - production mode on custom network

```
# Part 4.5 - production mode on a custom network
$ docker network create app-network
3b09eef2f29e9730fe37ec9459c74a7111b5bab8678877eda1e701c2cf1f057d

$ docker run -d --name prod-postgres --network app-network -e POSTGRES_USER=produser -e POSTGRES_PASSWORD=prodpass123 -e POS
TGRES_DB=proddb postgres:15-alpine
fa28b1755b35f5e7ddd31fa363faf6bfdcd7ff1f2cd24af2678114dfccc25953

# .env.production has DB_HOST=prod-postgres
$ grep -E ^(APP_ENV|DEBUG|DB_HOST)= .env.production
APP_ENV=production
DEBUG=false
DB_HOST=prod-postgres

$ docker run -d --name config-app-prod --network app-network --env-file .env.production -p 5001:5000 config-app
df9d7fd6650abfd63a71bca31d92f7cb71359a5a9fd31b6f9db8df6b48af6300

$ curl -s http://localhost:5001/
{"database_host":"prod-postgres","debug":false,"environment":"production","message":"Config API"}

$ curl -s http://localhost:5001/db-test
{"database":"PostgreSQL 15.19 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit","status":"success"}

$ docker ps --format table {{.Names}}\t{{.Image}}\t{{.Ports}}\t{{.Status}}
NAMES             IMAGE                PORTS                    STATUS
config-app-prod   config-app           0.0.0.0:5001->5000/tcp   Up 6 seconds
prod-postgres     postgres:15-alpine   5432/tcp                 Up 6 seconds
config-app-dev    config-app           0.0.0.0:5000->5000/tcp   Up 13 seconds
dev-postgres      postgres:15-alpine   0.0.0.0:5432->5432/tcp   Up 20 seconds
```

9. Docker Compose - Lab 5

- docker-compose.yml with 4 services: api (Flask), db (PostgreSQL 15), redis, nginx (optional reverse proxy).
- Named volumes postgres_data and redis_data; healthchecks for db, redis and api; depends_on with condition: service_healthy; variables from .env; custom networks backend and frontend; restart: unless-stopped.
- API endpoints tested: /, /health, /db-test, /cache-test, POST /users, GET /users, GET /users/<id> (first from the database, then from the Redis cache). Nginx proxies http://localhost:8080 to the API.
- docker compose down followed by up -d kept the data (users still present). .env is ignored by Git; .env.example is committed.

Docker Compose - Lab 5

docker compose up, all services healthy

Lab 5 - docker compose up, healthy services

```
$ docker compose config --quiet && echo "compose file is valid"
compose file is valid

$ docker compose config --services
redis
db
api
nginx

$ docker compose build api   # (sandbox: built via helper with CA build secret, same image name)
#11 naming to docker.io/library/lab5-compose-api:latest done

$ docker compose up -d
 Network lab5-compose_backend Created
 Container lab5-compose-db-1 Creating
 Container lab5-compose-redis-1 Creating
 Network lab5-compose_frontend Created
 Network lab5-compose_frontend Created
 Container lab5-compose-db-1 Created
 Container lab5-compose-redis-1 Created
 Container lab5-compose-api-1 Creating
 Container lab5-compose-api-1 Created
 Container lab5-compose-nginx-1 Creating
 Container lab5-compose-nginx-1 Created
 Container lab5-compose-redis-1 Starting
 Container lab5-compose-db-1 Starting
 Container lab5-compose-redis-1 Started
 Container lab5-compose-db-1 Started
 Container lab5-compose-redis-1 Waiting
 Container lab5-compose-db-1 Waiting
 Container lab5-compose-redis-1 Healthy
 Container lab5-compose-db-1 Healthy
 Container lab5-compose-api-1 Starting
 Container lab5-compose-api-1 Started
 Container lab5-compose-api-1 Waiting
 Container lab5-compose-api-1 Healthy
 Container lab5-compose-nginx-1 Starting
 Container lab5-compose-nginx-1 Started

$ docker compose ps
NAME                  IMAGE                COMMAND                  SERVICE   CREATED          STATUS                    PO
RTS
lab5-compose-api-1    lab5-compose-api     "python app.py"          api       27 seconds ago   Up 21 seconds (healthy)   0.
0.0.0:5000->5000/tcp
lab5-compose-db-1     postgres:15-alpine   "docker-entrypoint.s…"   db        28 seconds ago   Up 27 seconds (healthy)   54
32/tcp
lab5-compose-nginx-1  nginx:alpine         "/docker-entrypoint.…"   nginx     27 seconds ago   Up 15 seconds             0.
0.0.0:8080->80/tcp
lab5-compose-redis-1  redis:7-alpine       "docker-entrypoint.s…"   redis     28 seconds ago   Up 27 seconds (healthy)   63
79/tcp
```

Docker Compose - Lab 5

API endpoints, cache, Nginx

```
Lab 5 - API endpoints, Redis cache, Nginx proxy

$ curl -s localhost:5000/health
{"status":"healthy"}

$ curl -s localhost:5000/db-test
{"database":"PostgreSQL 15.19 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit","status":"success"}

$ curl -s localhost:5000/cache-test
{"redis":"connected","status":"success","test_value":"test_value"}

$ curl -s -X POST localhost:5000/users -H "Content-Type: application/json" -d '{"name": "Alice", "email": "alice@example.co
m"}'
{"email":"alice@example.com","id":1,"name":"Alice"}
$ curl -s -X POST localhost:5000/users ... Bob
{"email":"bob@example.com","id":2,"name":"Bob"}
$ curl -s -X POST localhost:5000/users ... duplicate email
{"error":"email already exists"}

$ curl -s localhost:5000/users
[{"email":"alice@example.com","id":1,"name":"Alice"},{"email":"bob@example.com","id":2,"name":"Bob"}]

$ curl -s localhost:5000/users/1   # 1st request -> database
{"source":"database","user":{"email":"alice@example.com","id":1,"name":"Alice"}}
$ curl -s localhost:5000/users/1   # 2nd request -> cache (Redis)
{"source":"cache","user":{"email":"alice@example.com","id":1,"name":"Alice"}}
$ docker compose exec redis redis-cli TTL user:1
60

$ curl -s localhost:8080/health   # via Nginx reverse proxy
{"status":"healthy"}
```

Docker Compose - Lab 5

Networks, DNS, volumes, persistence (part 1/2)

```
Lab 5 - network/DNS, volumes, persistence after down/up, .env ignored

# Part 4.3 - networks and DNS from inside the api container
$ docker compose exec api getent hosts db redis
172.18.0.3      db
172.18.0.2      redis

$ docker compose exec api python -c "socket.gethostbyname(...)"
db -> 172.18.0.3
redis -> 172.18.0.2

$ docker network ls | grep lab5
NETWORK ID     NAME                    DRIVER    SCOPE
8491903529f7   lab5-compose_backend    bridge    local
7858da7357fb   lab5-compose_frontend   bridge    local

$ docker volume ls | grep lab5
local     lab5-compose_postgres_data
local     lab5-compose_redis_data

$ docker compose exec api whoami
appuser

# Persistence: down (without -v) and up again, data must survive
$ docker compose down
 Container lab5-compose-db-1 Stopping
 Container lab5-compose-redis-1 Stopping
 Container lab5-compose-db-1 Stopped
 Container lab5-compose-db-1 Removing
 Container lab5-compose-db-1 Removed
 Container lab5-compose-redis-1 Stopped
 Container lab5-compose-redis-1 Removing
 Container lab5-compose-redis-1 Removed
 Network lab5-compose_frontend Removing
 Network lab5-compose_backend Removing
 Network lab5-compose_backend Removed
 Network lab5-compose_frontend Removed

$ docker compose up -d
 Container lab5-compose-redis-1 Started
 Container lab5-compose-db-1 Started
 Container lab5-compose-redis-1 Waiting
 Container lab5-compose-db-1 Waiting
 Container lab5-compose-redis-1 Healthy
 Container lab5-compose-db-1 Healthy
 Container lab5-compose-api-1 Starting
 Container lab5-compose-api-1 Started
 Container lab5-compose-api-1 Waiting
 Container lab5-compose-api-1 Healthy
 Container lab5-compose-nginx-1 Starting
 Container lab5-compose-nginx-1 Started

$ docker compose ps
NAME                  IMAGE                COMMAND                  SERVICE   CREATED          STATUS                    PO
RTS
lab5-compose-api-1    lab5-compose-api     "python app.py"          api       21 seconds ago   Up 15 seconds (healthy)   0.
0.0.0:5000->5000/tcp
lab5-compose-db-1     postgres:15-alpine   "docker-entrypoint.s…"   db        21 seconds ago   Up 21 seconds (healthy)   54
32/tcp
lab5-compose-nginx-1  nginx:alpine         "/docker-entrypoint.…"   nginx     21 seconds ago   Up 10 seconds             0.
0.0.0:8080->80/tcp
lab5-compose-redis-1  redis:7-alpine       "docker-entrypoint.s…"   redis     21 seconds ago   Up 21 seconds (healthy)   63
79/tcp

$ curl -s localhost:5000/users   # users still there
[{"email":"alice@example.com","id":1,"name":"Alice"},{"email":"bob@example.com","id":2,"name":"Bob"}]

# .env is not in Git:
$ git check-ignore -v .env
docker-lab/practice/lab5-compose/.gitignore:1:.env      .env

$ docker compose logs api | tail -n 8
api-1  |  * Running on all addresses (0.0.0.0)
api-1  |  * Running on http://127.0.0.1:5000
api-1  |  * Running on http://172.18.0.4:5000
api-1  | Press CTRL+C to quit
api-1  | 127.0.0.1 - - [04/Nov/2026 06:26:30] "GET /health HTTP/1.1" 200 -
api-1  | 127.0.0.1 - - [04/Nov/2026 06:26:35] "GET /health HTTP/1.1" 200 -
```

Docker Compose - Lab 5

Networks, DNS, volumes, persistence (part 2/2)

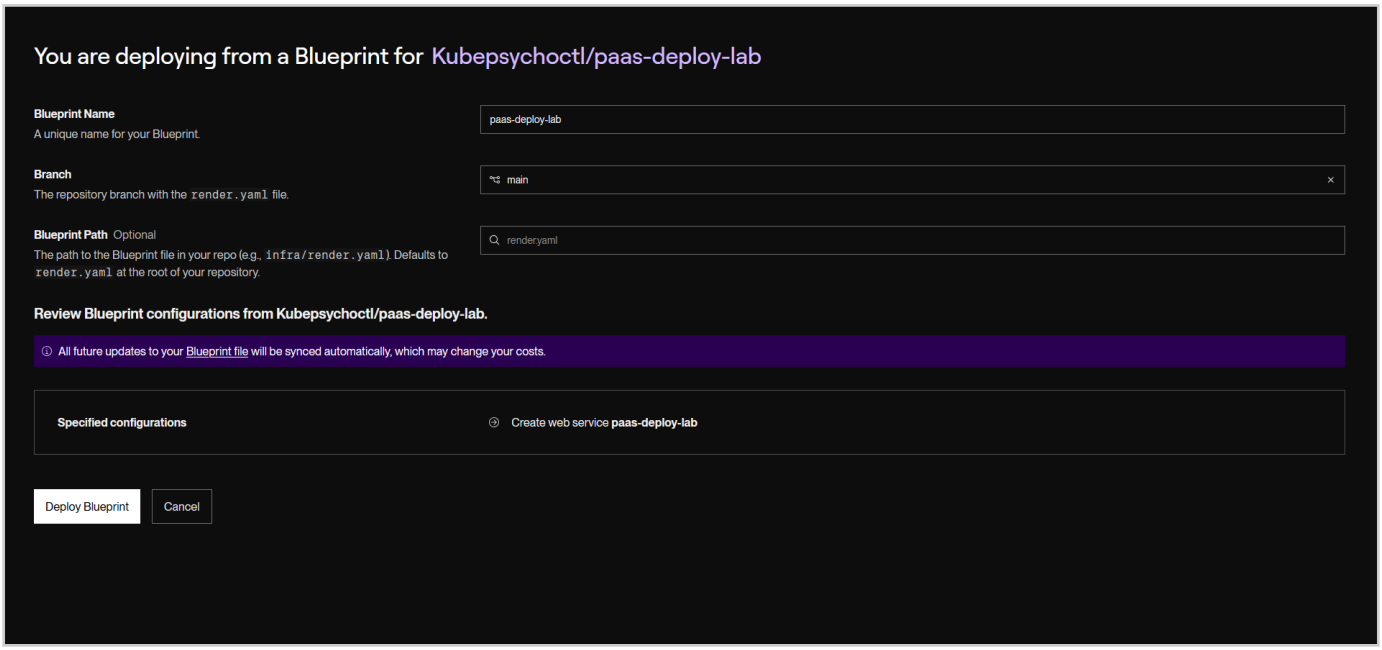
```
api-1  | 127.0.0.1 - - [04/Nov/2026 06:26:40] "GET /health HTTP/1.1" 200 -
api-1  | 172.18.0.1 - - [04/Nov/2026 06:26:40] "GET /users HTTP/1.1" 200 -
```

10. Part 10 - PaaS deployment (Render)

- Platform: Render (free web service). Public HTTPS URL: https://paas-deploy-lab.onrender.com - /health returns HTTP 200 {"status":"healthy"}.
- Application: standalone Flask API (/, /health, /users) deployed with gunicorn. Code and CI: https://github.com/Kubepsychoctl/paas-deploy-lab (branch main). CI = GitHub Actions workflow .github/workflows/ci.yml running pytest.
- Automatic deploy on push to main: the Render service is linked to branch main of the repository (created from render.yaml with a Blueprint); every push to main triggers a deploy attempt.
- Setting used for 'deploy only if CI passes': Auto-Deploy = 'After CI Checks Pass' (dashboard: service Settings, Auto-Deploy). In render.yaml it is autoDeployTrigger: checksPass.
- Where found in the documentation: https://render.com/docs/deploys, section 'Configuring auto-deploys' (the option list: On Commit / After CI Checks Pass / Off) and section 'Integrating with CI' (Render does not deploy if a check fails or if zero checks are detected; GitHub Actions checks are supported). The render.yaml key autoDeployTrigger (commit | checksPass | off) is in https://render.com/docs/blueprint-spec. Free tier limits: https://render.com/docs/free.
- Proof: commit 667e131 (failing test, version v2-broken) - CI failed: https://github.com/Kubepsychoctl/paas-deploy-lab/actions/runs/42165416320 - the site kept serving v1 for the 3 minutes it was watched. Commit fad60dc (fix, version v3-fixed) - CI passed: https://github.com/Kubepsychoctl/paas-deploy-lab/actions/runs/42165777243 - the site switched to v3-fixed about 70 seconds later. First commit 2b91b0b (CI passed) was the initial deploy.
- Note: the free instance spins down after 15 minutes without traffic, so the first request after a pause can take about a minute.

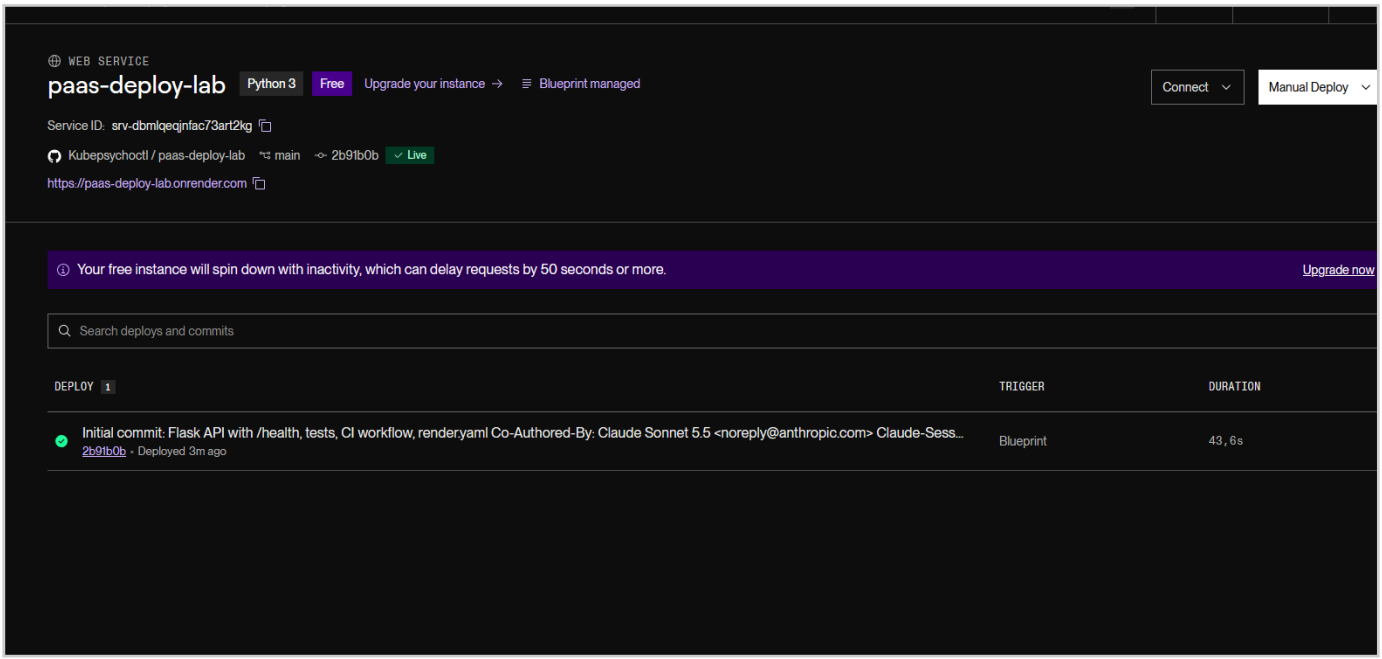
Part 10 - Render

Render: deploying the Blueprint for Kubepsychoctl/paas-deploy-lab (branch main)



Part 10 - Render

Render service page: Live, free plan, first deploy from commit 2b91b0b



Part 10 - Render

CI results of the three commits

```
Part 10 - CI results of the three commits pushed to main

$ gh api repos/Kubepsychoctl/paas-deploy-lab/actions/runs   # CI (GitHub Actions) result per commit on main
commit   CI result message
2b91b0b  success   Initial commit: Flask API with /health, tests, CI workflow, render.yaml
         https://github.com/Kubepsychoctl/paas-deploy-lab/actions/runs/42164796722
667e131  failure   Break /health (failing test): must NOT be deployed
         https://github.com/Kubepsychoctl/paas-deploy-lab/actions/runs/42165416320
fad60dc  success   Fix /health: tests pass again, deploy v3-fixed
         https://github.com/Kubepsychoctl/paas-deploy-lab/actions/runs/42165777243

# Render is set to 'After CI Checks Pass' (autoDeployTrigger: checksPass):
#   667e131 -> CI failure  -> Render does NOT deploy
#   fad60dc -> CI success  -> Render deploys
```

Part 10 - Render

Failing commit is not deployed (site stays on v1)

```
Part 10 - the commit with a failing test is NOT deployed

# 667e131 pushed 04:56:18 UTC (code says VERSION=v2-broken, /health returns 'unhealthy') -> CI FAILED at 04:56:45
# The live site keeps serving the previous version v1 (checked every ~18 s for 3 minutes):
$ curl -s https://paas-deploy-lab.onrender.com/ ; curl -s https://paas-deploy-lab.onrender.com/health
04:56:53  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:57:12  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:57:31  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:57:50  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:58:09  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:58:28  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:58:47  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:59:06  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:59:25  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
04:59:44  {"app":"PaaS Lab API","status":"running","version":"v1"} | health: {"status":"healthy"}
```

Part 10 - Render

Fix is deployed after CI passes (v3-fixed)

```
Part 10 - the fix is deployed after CI passes

# fad60dc pushed 05:00:14 UTC (fix, VERSION=v3-fixed) -> CI SUCCESS at 05:00:40 -> Render deploys
$ curl -s https://paas-deploy-lab.onrender.com/   # polled every 10 s
05:00:40 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:00:51 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:01:01 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:01:11 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:01:22 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:01:32 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:01:42 {"app":"PaaS Lab API","status":"running","version":"v1"}
05:01:53 {"app":"PaaS Lab API","status":"running","version":"v3-fixed"}

$ curl -s -o /dev/null -w 'HTTP %{http_code}' https://paas-deploy-lab.onrender.com/health
HTTP 200
```

Appendix - Git lab local simulation

Appendix

Step 1 (local simulation)

```
Step 1 - Create repository with README structure (Owner: Alice)

@team-docs$ git init --bare -b main /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origi
n.git
Initialized empty Git repository in /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origi
n.git/
@team-docs$ git clone /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git team-docs
-devops-team
Cloning into 'team-docs-devops-team'...
warning: You appear to have cloned an empty repository.
done.
alice@team-docs$ git config user.name "Alice" && git config user.email "alice@example.com"
alice@team-docs$ git add README.md && git commit -m 'Initial commit' && git push -u origin main
[main (root-commit) 968a7a8] Initial commit
 1 file changed, 17 insertions(+)
 create mode 100644 README.md
branch 'main' set up to track 'origin/main'.
fatal: expected 'acknowledgments', received 'packfile'
warning: push negotiation failed; proceeding anyway with push
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      main -> main
alice@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

List of team members here...

## Technology Stack

Technologies here...

## Setup Instructions

Steps to run the project here...
```

Appendix

Step 2 (local simulation)

```
Step 2 - Clone and create personal branches

@team-docs$ git clone /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git team-docs
-devops-team
Cloning into 'team-docs-devops-team'...
done.
bob@team-docs$ git config user.name "Bob" && git config user.email "bob@example.com"
@team-docs$ git clone /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git team-docs
-devops-team
Cloning into 'team-docs-devops-team'...
done.
carol@team-docs$ git config user.name "Carol" && git config user.email "carol@example.com"
alice@team-docs$ git checkout -b feature/alice
Switched to a new branch 'feature/alice'
alice@team-docs$ git branch
* feature/alice
  main
bob@team-docs$ git checkout -b feature/bob
Switched to a new branch 'feature/bob'
bob@team-docs$ git branch
* feature/bob
  main
carol@team-docs$ git checkout -b feature/carol
Switched to a new branch 'feature/carol'
carol@team-docs$ git branch
* feature/carol
  main
```

Appendix

Step 3 (local simulation)

```
Step 3 - Each member edits the SAME README.md

alice@team-docs$ git diff
diff --git a/README.md b/README.md
index 9a652b0..404f226 100644
--- a/README.md
+++ b/README.md
@@ -6,12 +6,15 @@ General description here...

 ## Team

-List of team members here...
+- Alice - DevOps Lead

 ## Technology Stack

-Technologies here...
+- Docker
+- Kubernetes

 ## Setup Instructions

-Steps to run the project here...
+1. Install Docker
+2. Run docker compose up
+3. Open localhost:8080
bob@team-docs$ git diff
diff --git a/README.md b/README.md
index 9a652b0..bc9ee1d 100644
--- a/README.md
+++ b/README.md
@@ -6,12 +6,15 @@ General description here...

 ## Team

-List of team members here...
+- Bob - CI/CD Engineer

 ## Technology Stack

-Technologies here...
+- Jenkins
+- Linux

 ## Setup Instructions

-Steps to run the project here...
+1. Install Jenkins
+2. Configure pipeline
+3. Run first build
carol@team-docs$ git diff
diff --git a/README.md b/README.md
index 9a652b0..e83e40e 100644
--- a/README.md
+++ b/README.md
@@ -6,12 +6,15 @@ General description here...

 ## Team

-List of team members here...
+- Carol - Cloud Engineer

 ## Technology Stack

-Technologies here...
+- Terraform
+- AWS

 ## Setup Instructions

-Steps to run the project here...
+1. Install Terraform
+2. Run terraform init
+3. Run terraform apply
```

Appendix

Step 4 (local simulation)

```
Step 4 - Commit and push

alice@team-docs$ git status
On branch feature/alice
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
alice@team-docs$ git add README.md
alice@team-docs$ git commit -m "Add Alice: team info and setup instructions"
[feature/alice 72eb7b6] Add Alice: team info and setup instructions
 1 file changed, 6 insertions(+), 3 deletions(-)
alice@team-docs$ git push origin feature/alice
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      feature/alice -> feature/alice
bob@team-docs$ git status
On branch feature/bob
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
bob@team-docs$ git add README.md
bob@team-docs$ git commit -m "Add Bob: team info and setup instructions"
[feature/bob 919f4a7] Add Bob: team info and setup instructions
 1 file changed, 6 insertions(+), 3 deletions(-)
bob@team-docs$ git push origin feature/bob
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      feature/bob -> feature/bob
carol@team-docs$ git status
On branch feature/carol
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
carol@team-docs$ git add README.md
carol@team-docs$ git commit -m "Add Carol: team info and setup instructions"
[feature/carol b91eb59] Add Carol: team info and setup instructions
 1 file changed, 6 insertions(+), 3 deletions(-)
carol@team-docs$ git push origin feature/carol
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      feature/carol -> feature/carol
```

Appendix

Step 5 (local simulation)

```
Step 5 - Pull Requests (simulated; no GitHub UI available)

On GitHub each branch gets a PR titled 'Add [name]: team contribution'. Here PR = merge of the branch into main with a 'Merg
e pull request #N' commit.
alice@team-docs$ git branch -r
  origin/feature/alice
  origin/main
```

Appendix

Step 6 - Bob's conflict (local simulation) (part 1/2)

```
Step 6 - Merge PR #1, then conflict on PR #2

alice@team-docs$ git fetch origin
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * [new branch]      feature/bob   -> origin/feature/bob
 * [new branch]      feature/carol -> origin/feature/carol
alice@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
alice@team-docs$ git checkout main && git pull origin main
Your branch is up to date with 'origin/main'.
Already up to date.
Already on 'main'
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
alice@team-docs$ git merge --no-ff feature/alice -m "Merge pull request #1 - feature/alice" && git push origin main
Merge made by the 'ort' strategy.
 README.md | 9 ++++++---
 1 file changed, 6 insertions(+), 3 deletions(-)
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   968a7a8..feb6df7  main -> main
alice@team-docs$ git merge --no-ff origin/feature/bob -m 'Merge pull request #2 - feature/bob'
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
GitHub would show: 'This branch has conflicts that must be resolved' -> abort and let Bob fix it on his branch.
alice@team-docs$ git merge --abort
bob@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
bob@team-docs$ git pull origin main
Updating 968a7a8..feb6df7
Fast-forward
 README.md | 9 ++++++---
 1 file changed, 6 insertions(+), 3 deletions(-)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   968a7a8..feb6df7  main       -> origin/main
bob@team-docs$ git checkout feature/bob
Switched to branch 'feature/bob'
bob@team-docs$ git merge main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
bob@team-docs$ git status
On branch feature/bob
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
bob@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

<<<<<<< HEAD
- Bob - CI/CD Engineer

## Technology Stack

- Jenkins
- Linux

## Setup Instructions

1. Install Jenkins
2. Configure pipeline
3. Run first build
=======
```

Appendix

Step 6 - Bob's conflict (local simulation) (part 2/2)

```
- Alice - DevOps Lead

## Technology Stack

- Docker
- Kubernetes

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
>>>>>>> main
bob@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
bob@team-docs$ grep -nE '^(<<<<<<<|=======|>>>>>>>)' README.md || echo 'no conflict markers left'
no conflict markers left
bob@team-docs$ git add README.md
bob@team-docs$ git commit -m "Resolve merge conflict: combine team contributions"
[feature/bob 54e32db] Resolve merge conflict: combine team contributions
bob@team-docs$ git push origin feature/bob
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   919f4a7..54e32db  feature/bob -> feature/bob
alice@team-docs$ git checkout main && git pull origin main && git fetch origin
Your branch is up to date with 'origin/main'.
Already up to date.
Already on 'main'
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
   919f4a7..54e32db  feature/bob -> origin/feature/bob
alice@team-docs$ git merge --no-ff origin/feature/bob -m 'Merge pull request #2 - feature/bob' && git push origin main
Merge made by the 'ort' strategy.
 README.md | 6 ++++++
 1 file changed, 6 insertions(+)
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   feb6df7..df7ae91  main -> main
```

Appendix

Step 6 - Carol's conflict (local simulation) (part 1/2)

```
Step 6b - Same conflict for PR #3 (Carol)

carol@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
carol@team-docs$ git pull origin main
Updating 968a7a8..df7ae91
Fast-forward
 README.md | 15 ++++++++++---
 1 file changed, 12 insertions(+), 3 deletions(-)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   968a7a8..df7ae91  main       -> origin/main
carol@team-docs$ git checkout feature/carol
Switched to branch 'feature/carol'
carol@team-docs$ git merge main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
carol@team-docs$ git status
On branch feature/carol
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
carol@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer
- Carol - Cloud Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux
- Terraform
- AWS

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
7. Install Terraform
8. Run terraform init
9. Run terraform apply
carol@team-docs$ git add README.md
carol@team-docs$ git commit -m "Resolve merge conflict: combine team contributions"
[feature/carol 585cf10] Resolve merge conflict: combine team contributions
carol@team-docs$ git push origin feature/carol
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   b91eb59..585cf10  feature/carol -> feature/carol
alice@team-docs$ git checkout main && git pull origin main && git fetch origin
Your branch is up to date with 'origin/main'.
Already up to date.
Already on 'main'
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
   b91eb59..585cf10  feature/carol -> origin/feature/carol
alice@team-docs$ git merge --no-ff origin/feature/carol -m 'Merge pull request #3 - feature/carol' && git push origin main
```

Appendix

Step 6 - Carol's conflict (local simulation) (part 2/2)

```
Merge made by the 'ort' strategy.
 README.md | 6 ++++++
 1 file changed, 6 insertions(+)
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   df7ae91..8b66926  main -> main
```

Appendix

Step 7 - final history (local simulation) (part 1/2)

```
Step 7 - Sync and final history

alice@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Already on 'main'
alice@team-docs$ git pull origin main
Already up to date.
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
bob@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
bob@team-docs$ git pull origin main
Updating feb6df7..8b66926
Fast-forward
 README.md | 12 ++++++++++++
 1 file changed, 12 insertions(+)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   feb6df7..8b66926  main       -> origin/main
carol@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
carol@team-docs$ git pull origin main
Updating df7ae91..8b66926
Fast-forward
 README.md | 6 ++++++
 1 file changed, 6 insertions(+)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   df7ae91..8b66926  main       -> origin/main
alice@team-docs$ git log --oneline --graph --all
*   8b66926 Merge pull request #3 - feature/carol
|\
| *   585cf10 Resolve merge conflict: combine team contributions
| |\
| |/
|/|
* |   df7ae91 Merge pull request #2 - feature/bob
|\ \
| * \   54e32db Resolve merge conflict: combine team contributions
| |\ \
| |/ /
|/| |
* | |   feb6df7 Merge pull request #1 - feature/alice
|\ \ \
| * | | 72eb7b6 Add Alice: team info and setup instructions
|/ / /
| * / 919f4a7 Add Bob: team info and setup instructions
|/ /
| * b91eb59 Add Carol: team info and setup instructions
|/
* 968a7a8 Initial commit
alice@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer
- Carol - Cloud Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux
- Terraform
- AWS

## Setup Instructions

1. Install Docker
```

Appendix

Step 7 - final history (local simulation) (part 2/2)

```
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
7. Install Terraform
8. Run terraform init
9. Run terraform apply
```